

Synopsis:

Project Title: TEXT PURITY – Plagiarism Detection System

Introduction:

In the present digital world, copying and reusing others' work has become very common due to the easy availability of online content. This has increased cases of plagiarism in academic and professional writing. Many existing plagiarism detection tools like Turnitin or Grammarly are effective but require online access, which can risk the privacy of users' data.

To overcome this problem, **Text Purity – Plagiarism Detection System** has been developed. It is an **offline tool** designed to detect **copied, paraphrased, and AI-generated text** while maintaining complete **data security and privacy**. The system is built using **Python and Flask**, with integrated **machine learning** and **NLP techniques** for accurate and intelligent plagiarism detection.

Objectives:

1. To develop a plagiarism detection system that identifies copied, rephrased, or AI-generated text.
2. To ensure complete data privacy by operating entirely offline.
3. To support multiple file formats like .txt, .pdf, .docx, and image files using OCR.
4. To apply NLP techniques such as **TF-IDF** and **Cosine Similarity** for similarity detection.
5. To provide a simple and user-friendly web interface for document upload and report viewing.

Scope:

The system is suitable for **students, teachers, researchers, and professionals** who want to check their documents for originality. It works offline, ensuring that sensitive or confidential data remains secure. The tool can analyze text from different file types, including scanned or handwritten documents, and highlight plagiarized or AI-generated sections.

Technology Used:

- **Language:** Python 3.12+
- **Framework:** Flask
- **Libraries:** scikit-learn, spaCy, NLTK, Transformers (BERT/SBERT), PyTesseract (OCR)
- **Database:** SQLite
- **Frontend:** HTML, CSS, JavaScript

Expected Outcome:

The project will produce a secure and efficient plagiarism detection system that:

- Works completely offline to protect user data.
- Accurately detects copied, paraphrased, and AI-generated text.
- Provides easy-to-understand plagiarism reports.
- Encourages originality and academic honesty.

Conclusion:

The **Text Purity – Plagiarism Detection System** offers a reliable, privacy-focused solution to modern plagiarism challenges. By combining machine learning, natural language processing, and OCR, it provides a comprehensive way to verify originality while ensuring complete data security.